CURIONEST

Polyprotic Acids & Mixed Solutions

Practice Pack · Stepwise Ionization · Moles-First Stoichiometry · Full Answer Key

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

What This Pack Covers

1. Writing stepwise ionization equations for polyprotic acids
2. Using K_{a1} vs K_{a2} to calculate $[H_3O^+]$ for diprotic acids
3. Applying the moles-first method to mixed-solution problems
4. Identifying the limiting reactant and leftover species
5. Determining which species dominates at a given pH

Reading Passage — Polyprotic Acids and Mixed Solutions

Polyprotic acids donate protons one at a time. For carbonic acid: $H_2CO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$ with $K_{a1} = 4.3 \times 10^{-7}$, then $HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CO_3^{2-}(aq)$ with $K_{a2} = 4.7 \times 10^{-11}$. Each step is harder because a proton must be pulled from a negatively charged ion, so $K_{a1} >> K_{a2}$.

Because K_{a1} dominates, $[H_3O^+]$ for a diprotic acid is essentially set by the first ionization step alone. The second step contributes negligibly.

For mixed-solution problems, use the moles-first method: (1) convert every solute to moles, (2) react to completion — identify the limiting reactant and the leftover, (3) divide by the TOTAL volume, (4) calculate pH from the remaining strong acid, strong base, or weak conjugate pair.

The intermediate ion of a polyprotic acid (HCO_3^-, HSO_4^-, $H_2PO_4^-$) is amphiprotic and buffers the solution between the two equivalence points of a titration.

★ **Common mistake:** Writing both ionization steps in a single equation — polyprotic acids ionize one proton at a time.

★ **Common mistake:** Dividing by one solution's volume instead of the total volume after mixing.

★ **Common mistake:** Forgetting to check which reactant is the limiting reactant.

Quick Check — Multiple Choice (Questions 1–8)

1. A polyprotic acid ionizes:
   (A) All protons at once
   (B) One proton at a time
   (C) Only in base
   (D) Never in water

2. For a diprotic acid, K_{a1} is always _____ K_{a2}.
   (A) Less than
   (B) Equal to
   (C) Greater than
   (D) Unrelated to

3. In a 0.050 M solution of H_2S, which species has the highest concentration?
   (A) S^{2-}
   (B) HS^-
   (C) H_3O^+
   (D) H_2S

4. 50.0 mL of 0.100 M HCl mixes with 50.0 mL of 0.100 M NaOH. The resulting solution has pH:
   (A) 0
   (B) 7.00
   (C) 14
   (D) 1.00

5. 25.0 mL of 0.200 M HCl mixes with 75.0 mL of 0.050 M NaOH. The excess species is:
   (A) NaOH (0.00125 mol)
   (B) HCl (0.00125 mol)
   (C) HCl (0.00500 mol)

(D) NaOH (0.00375 mol)

6. At pH 5.0, the dominant phosphate species is (pKa1 = 2.15, pKa2 = 7.20):
(A) H_3PO_4
(B) $H_2PO_4^-$
(C) HPO_4^{2-}
(D) PO_4^{3-}

7. 30.0 mL of 0.100 M H_2SO_4 (diprotic, fully ionizing) provides how many moles of H^+?
(A) 0.00300
(B) 0.00600
(C) 0.00150
(D) 0.0300

8. The intermediate ion HCO_3^- is best described as:
(A) A strong acid
(B) A strong base
(C) Amphiprotic
(D) Inert

Calculate — Show Your Work (Questions 9–14)

9. Calculate $[H_3O^+]$ and pH for a 0.010 M solution of H_2S ($K_{a1} = 1.0 \times 10^{-7}$).

10. Calculate $[H_3O^+]$ and pH for a 0.134 M solution of H_2CO_3 ($K_{a1} = 4.3 \times 10^{-7}$).

11. 50.0 mL of 0.100 M HCl is mixed with 50.0 mL of 0.060 M NaOH. Calculate the pH.

12. 40.0 mL of 0.150 M KOH is mixed with 60.0 mL of 0.080 M HCl. Calculate the pH.

13. 30.0 mL of 0.100 M H_2SO_4 is mixed with 70.0 mL of 0.050 M NaOH. Calculate the pH.

14. 100.0 mL of 0.050 M HCN ($K_a = 6.2 \times 10^{-10}$) is mixed with 50.0 mL of 0.100 M NaOH. Calculate the pH.

Short Answer & Think (Questions 15–18)

15. Why does the second ionization of a diprotic acid contribute almost nothing to $[H_3O^+]$?

16. A student mixes 50 mL of 0.10 M HCl with 50 mL of 0.10 M NaOH and reports pH = 1. Explain the error.

17. Explain the moles-first method and why the total volume matters in mixed-solution calculations.

18. For phosphoric acid (pK_{a1} = 2.15, pK_{a2} = 7.20, pK_{a3} = 12.35), which species dominates at pH 1.0, 5.0, 9.0, and 13.0?

Key Terms

Term	Definition
polyprotic acid	Acid with more than one ionizable proton (H_2CO_3, H_3PO_4)
stepwise ionization	Protons removed one at a time, each with its own K_a
amphiprotic intermediate	Ion between steps (HCO_3^-) that can act as acid or base
moles-first method	Convert to moles → react to completion → divide by total volume
limiting reactant	The reactant used up first in a reaction
species distribution	Which protonated form dominates at a given pH

Answer Key

All answers with explanations are embedded in the questions above — use the practice sections for the complete key.

Quick Check — Multiple Choice (Questions 1–8)

1. One proton at a time (stepwise ionization)

Why: Each step has its own Ka; the first step dominates $[H_3O^+]$.

2. Greater than (it is harder to remove a proton from a negative ion)

Why: Electrostatic attraction makes the second proton harder to remove.

3. H_2S (the acid itself — weak acids ionize only slightly)

Why: With $K_{a1} = 1.0 \times 10^{-7}$, only ~0.1% ionizes, so H_2S remains by far the most abundant species.

4. 7.00 (equal moles of strong acid and base exactly neutralize)

Why: Moles HCl = moles NaOH = 0.00500; complete neutralization leaves pure water with the salt.

5. HCl (0.00125 mol)

Why: Moles HCl = 0.00500, moles NaOH = 0.00375; leftover acid = 0.00125 mol in 0.100 L.

6. $H_2PO_4^-$ (pH between pK_{a1} and pK_{a2})

Why: Between the two pKa values, the singly deprotonated form dominates.

7. 0.00600 (0.0300 L × 0.100 M × 2 protons)

Why: Each H_2SO_4 molecule supplies two H^+; multiply moles of acid by 2.

8. Amphiprotic (can act as acid or base)

Why: HCO_3^- can donate a proton (→ CO_3^{2-}) or accept one (→ H_2CO_3).

Calculate — Show Your Work (Questions 9–14)

9. $x^2/0.010 = 1.0 \times 10^{-7} \rightarrow x = 3.2 \times 10^{-5}$ M; pH = 4.50

10. $x^2/0.134 = 4.3 \times 10^{-7} \rightarrow x = 2.4 \times 10^{-4}$ M; pH = 3.62

11. Moles HCl = 0.00500; NaOH = 0.00300; excess H^+ = 0.00200 mol in 0.100 L → $[H_3O^+]$ = 0.0200 M; pH = 1.70

12. Moles KOH = 0.00600; HCl = 0.00480; excess OH^- = 0.00120 mol in 0.100 L → $[OH^-]$ = 0.0120 M; pOH = 1.92; pH = 12.08

13. H^+ = 0.00600 mol; OH^- = 0.00350 mol; excess H^+ = 0.00250 mol in 0.100 L → $[H_3O^+]$ = 0.0250 M; pH = 1.60

14. Exactly neutralized → $[CN^-]$ = 0.00500/0.150 = 0.0333 M. $K_b = 1.6 \times 10^{-5}$; $[OH^-] = 7.3 \times 10^{-4}$ M; pOH = 3.14; pH = 10.86

Short Answer & Think (Questions 15–18)

15. K_{a2} is typically 10^4–10^5 times smaller than K_{a1} because removing a proton from a negatively charged ion is much harder; the tiny $[HS^-]$-type intermediate ionizes barely at all, so its H_3O^+ contribution is negligible.

16. Equal moles of strong acid and strong base exactly neutralize: 0.0050 mol each, no leftover. The result is neutral salt water, pH = 7 — the student likely forgot to react the acid and base to completion.

17. Convert to moles, react to completion, then divide the leftover moles by the combined (total) volume to get concentration — pH depends on concentration, and the volume after mixing is the sum of both solutions.

18. pH 1.0: H_3PO_4 · pH 5.0: $H_2PO_4^-$ · pH 9.0: HPO_4^{2-} · pH 13.0: PO_4^{3-}

